



Data Science Internship at Unlox

Submitted by:

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Internship Overview

Organization

Unlox — student learning and career development ecosystem

Role

Data Science Intern

Focus Areas

Python, Data Analysis, SQL, Visualization, and Machine Learning



Learning Objectives

Programming

Python, NumPy, Pandas

Visualization

Matplotlib, Power BI

Database

SQL / MySQL

ML Library

Scikit-learn

Development

Jupyter, GitHub

Projects 1 & 2:

Foundational Skills

Project 1 — Python Calculator

Developed an arithmetic calculator with robust input validation, supporting six operations: addition, subtraction, multiplication, division, modulus, and exponentiation. Strengthened understanding of control flow, loops, and error handling.

Project 2 — SQL Student Database

Created a relational database managing student records and course information. Practiced core SQL operations including `CREATE`, `INSERT`, `SELECT`, `WHERE`, and `ORDER BY`, establishing a foundation for data preparation and retrieval workflows.

Project 3:

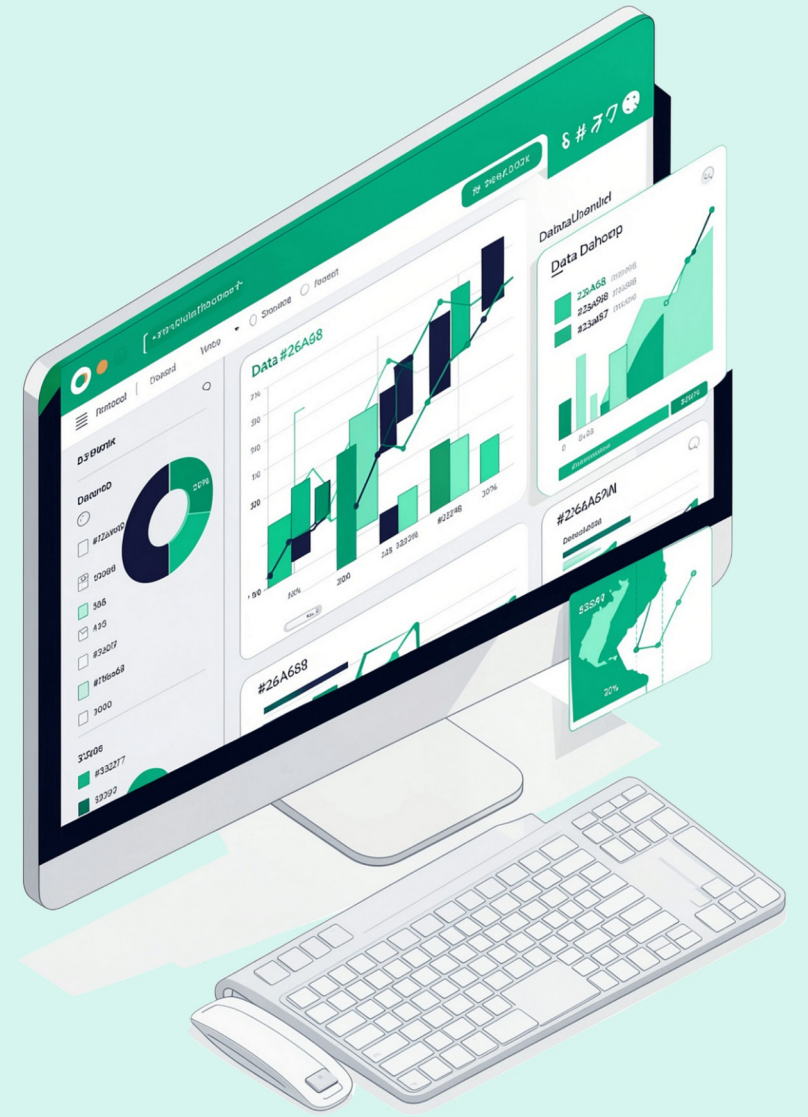
Excel & Power BI

Excel Analysis

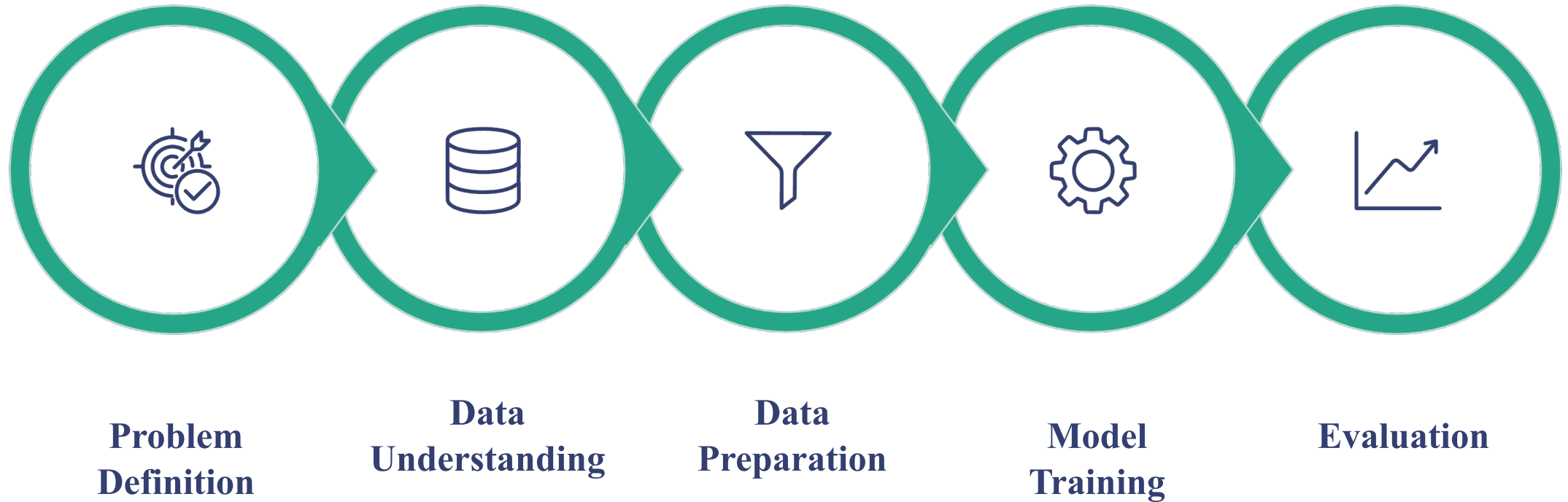
Applied formulas and conditional logic — IF, SUMIF — for salary calculations and data classification across structured datasets.

Power BI Dashboard

Built an interactive dashboard visualizing Airbnb booking and revenue data, developing skills in communicating analytical insights to non-technical audiences.



Machine Learning Workflow



This structured methodology was applied consistently across both the regression and classification projects to ensure reproducibility and rigour.

Project 4:

House Price Prediction

Regression Task

- **Problem:** Predict house prices from property features
- **Dataset:** 545 records, 13 features (area, bedrooms, bathrooms, amenities)
- **Model:** Linear Regression via Scikit-learn

0.456

R^2 Score

Model explains 45.6% of variance in house prices

545

Records

Training and test samples used



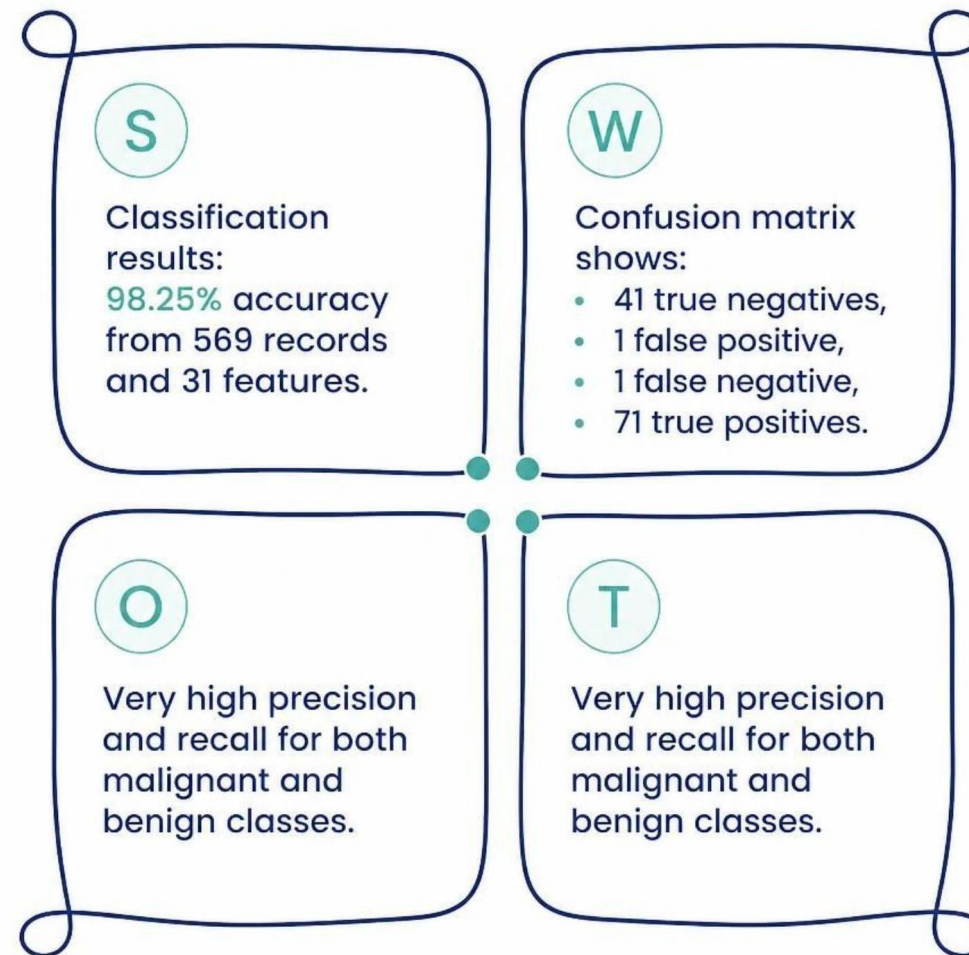
Project 5:

Breast Cancer Diagnosis

Classification Task

- **Problem:** Classify tumours as malignant or benign
- **Dataset:** 569 records, 31 computed cell-nucleus features
- **Model:** Binary classification with Scikit-learn
- **Confusion Matrix:** $\begin{bmatrix} 41 & 1 \\ 1 & 71 \end{bmatrix}$ — only 2 misclassifications

✓ Strong precision and recall alongside 98.25% accuracy confirm the model's clinical reliability.



Outcomes

98.25%

Classification Accuracy

Breast cancer diagnosis model

45.6%

Variance Explained

R^2 score — house price regression

5

Projects Completed

From calculator to ML classification

1114

Records Analysed

Across both ML datasets combined

Technical Skills Developed

01

Python & Data Manipulation

Programming fluency with NumPy and Pandas for real-world datasets

02

SQL Database Design

Relational schema design and structured query writing

03

Visualization & Dashboards

Matplotlib charts and interactive Power BI reporting

04

Machine Learning

Regression and classification with model evaluation and interpretation

